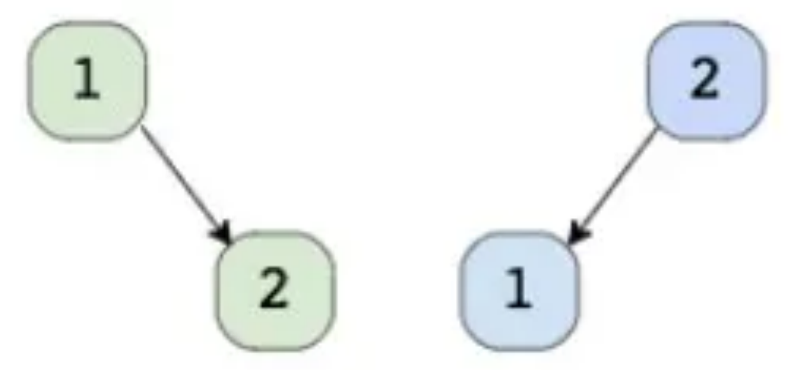
Solution: Problem Challenge 2: Structurally Unique Binary Search Trees

Problem Statement

Given a number `n`, write a function to return all structurally unique Binary Search Trees (BST) that can store values `1` to `n`?

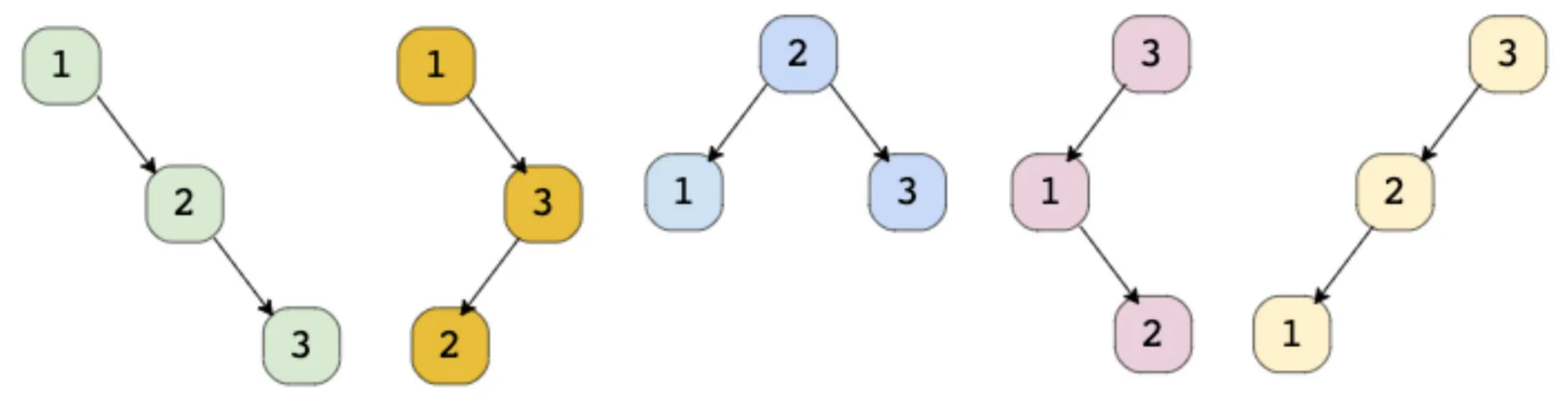
Example 1:

```
Input: 2
Output: List containing root nodes of all structurally unique BSTs.
Explanation: Here are the 2 structurally unique BSTs storing all numbers
from 1 to 2:
```



Example 2:

```
Input: 3
Output: List containing root nodes of all structurally unique BSTs.
Explanation: Here are the 5 structurally unique BSTs storing all numbers
from 1 to 3:
```

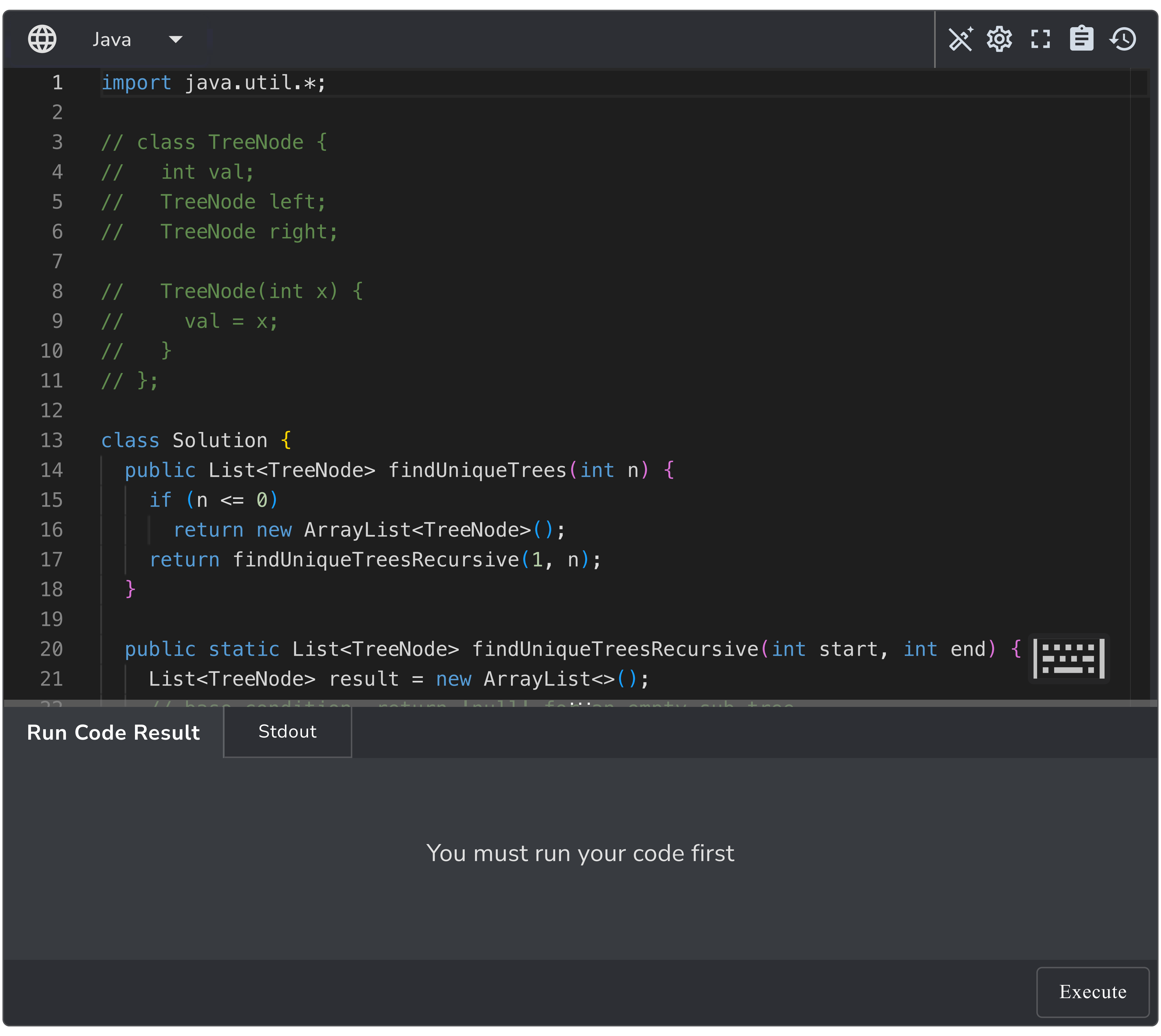


Solution

This problem follows the `Subsets` pattern and is quite similar to `Evaluate Expression`. Following a similar approach, we can iterate from 1 to 'n' and consider each number as the root of a tree. All smaller numbers will make up the left sub-tree and bigger numbers will make up the right sub-tree. We will make recursive calls for the left and right sub-trees

Code

Here is what our algorithm will look like:



```java
import java.util.*;

// class TreeNode {
//   int val;
//   TreeNode left;
//   TreeNode right;

//   TreeNode(int x) {
//     val = x;
//   }
// };

class Solution {
  public List<TreeNode> findUniqueTrees(int n) {
    if (n <= 0)
      return new ArrayList<TreeNode>();
    return findUniqueTreesRecursive(1, n);
  }

  public static List<TreeNode> findUniqueTreesRecursive(int start, int end) {
    List<TreeNode> result = new ArrayList<>();
```

You must run your code first

Time Complexity

The time complexity of this algorithm will be exponential and will be similar to Balanced Parentheses. Estimated time complexity will be $O(n*2^n)$ but the actual time complexity $(O(4^n/\sqrt{n})$ is bounded by the Catalan number and is beyond the scope of a coding interview. See more details here.

Space Complexity

The space complexity of this algorithm will be exponential too, estimated at $O(2^n)$, but the actual will be $(O(4^n/\sqrt{n})$.

Memoized version

Since our algorithm has overlapping subproblems, can we use memoization to improve it? We could, but every time we return the result of a subproblem from the cache, we have to clone the result list because these trees will be used as the left or right child of a tree. This cloning is equivalent to reconstructing the trees; therefore, the overall time complexity of the memoized algorithm will also be the same.